# Ani Le

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## EDUCATION

**University of Southern California,** Mechanical Engineering — January 2026 – December 2027

Related Coursework - Thermodynamics, Materials Behavior and Processing, Mechanics, Physics III, Computational Methods in Engineering, Mechoptronics, Dynamics, Computer Aided Engineering

Affiliations - Advanced Robotic Combat, Formula SAE Electric

## WORK EXPERIENCE

**USC Advanced Robotic Combat,** Drivetrain Co-Lead, Secretary — Los Angeles, CA | May 2026 – Present

- Designed and iterated on drivetrain systems for selecting motors, gearboxes, and testing different material wheel configurations to optimize speed, torque, and durability under impact conditions
- Collaborated with weapon, armor, and electronics subteams to ensure drivetrain integration met weight class constraints and competition requirements
- Coordinate the PLANTS program, providing 50 new students with hands-on training in CAD, soldering and engineering design

**DeepAware AI YC,** Hardware Engineering Intern — San Francisco, CA | June 2026 – August 2026

- Assembled, calibrated, and deployed OpenArm V1 and V2 bimanual robotic arms for enterprise customers at Robotics Center of Silicon Valley, an early-stage AI-integrated robotics startup
- Developed teleoperation functionality between leader and follower arms, integrating AI-driven control systems to enable precise, responsive real-time operation tailored to customer applications
- Served as technical liaison between vendors and customers, gathering requirements, troubleshooting design constraints, and ensuring final builds met customer specifications
- Represented the company at a Bay Area data summit, presenting OpenArm's capabilities and AI-powered teleoperation architecture to robotics and AI startups

**Bedrock Ocean Exploration,** Mechanical Engineering Intern — San Francisco, CA | May 2025 – August 2025

- Developed and tested biomimetic propulsion mechanisms for manta ray-inspired search and rescue robots, contributing to autonomous navigation systems designed to locate disaster survivors in aquatic and debris-filled environments
- Integrated multi-modal sensor arrays including thermal imaging and sonar detection systems into robotic prototypes, enabling real-time survivor identification during simulated disaster response scenarios
- Collaborated with cross-functional engineering teams to iterate on soft robotics designs, documenting performance metrics and design improvements that enhanced robot agility and operational efficiency in challenging terrain

**Liminal Insights,** Hardware Engineer Intern — Emeryville, CA | June 2024 – August 2024

- Performed mechanical design, engineering documentation, rapid prototyping, and simulation for components/subassemblies used in semi-automated processes
- Conducted build, assembly, testing, measurement variation analysis (ANOVA), and Root Cause Analysis (RCA) for system validation and process improvements
- Utilized CAD and Design for Excellence (DFX) methodologies, rapid prototyping techniques, manufacturing processes, and file management tools

**Oakland Lead,** Superposition — Oakland, CA | March 2021 – January 2026

- Founded the inaugural Superposition Oakland chapter, increasing access to technology education for underserved youth. Designed and taught coding curriculum to 50+ student participants, empowering the next generation of engineers and computer scientists
- Led planning and execution of 3 hackathons, providing opportunities for girls to showcase technical skills and build community. Hackathons drew 100+ attendees from across the Bay Area
- Launched R2 DGirls, an introductory robotics program for middle school girls at Edna Brewer Middle School. Developed curriculum focused on sparking passion for STEM fields through engaging hands-on activities with robots and circuits. R2 DGirls gained recognition as a model program and expanded to serve 150 students annually

## SKILLS

**Technical** — SolidWorks, Computer Aided Manufacturing, Siemens NX, CNC Machining, Hand Tools, CAD Design, Design for Excellence (DFX), Finite Element Analysis (FEA), Sensor Integration, Manufacturing Processes, Hydraulic Press Broaching, Bandsaw, Lathe, Mill, Hand Drill

**Soft Skills** — Technical Education & Training, Curriculum Design, Team Leadership, Cross-functional Collaboration, Event Planning & Execution, New Member Retention and Logistics

## AWARDS

**BattleBots Judges Decision Award,** Battlebots — May 2026

- Secured ARC's first official BattleBots victory via judges' decision against Superheavyweight Witch Doctor (325-lb) at BattleBots: Proving Grounds 2026, after a self-selected higher-difficulty matchup and a full electronics/drivetrain redesign completed in the weeks following the club's Pro League debut

**First Tech Competition Quarterfinals,** First Tech Competition — December 2024

- Advanced to Quarterfinals in the First Tech Competition, leading a team of six in developing an innovative robotics solution that excelled in autonomous navigation and precision control
- Engineered custom mechanical systems and programmed advanced pathfinding algorithms that enabled the robot to efficiently navigate obstacles and manipulate game elements, ultimately ranking in the top 8 teams among 32 competitors.